

GlowAI

SKIN CONDITION CLASSIFICATION FROM SYNTHETIC DATA

Submitted by: Lin Schneider, Reut Sasson, Ron Noiman | Course: GenAI | Year: 2026

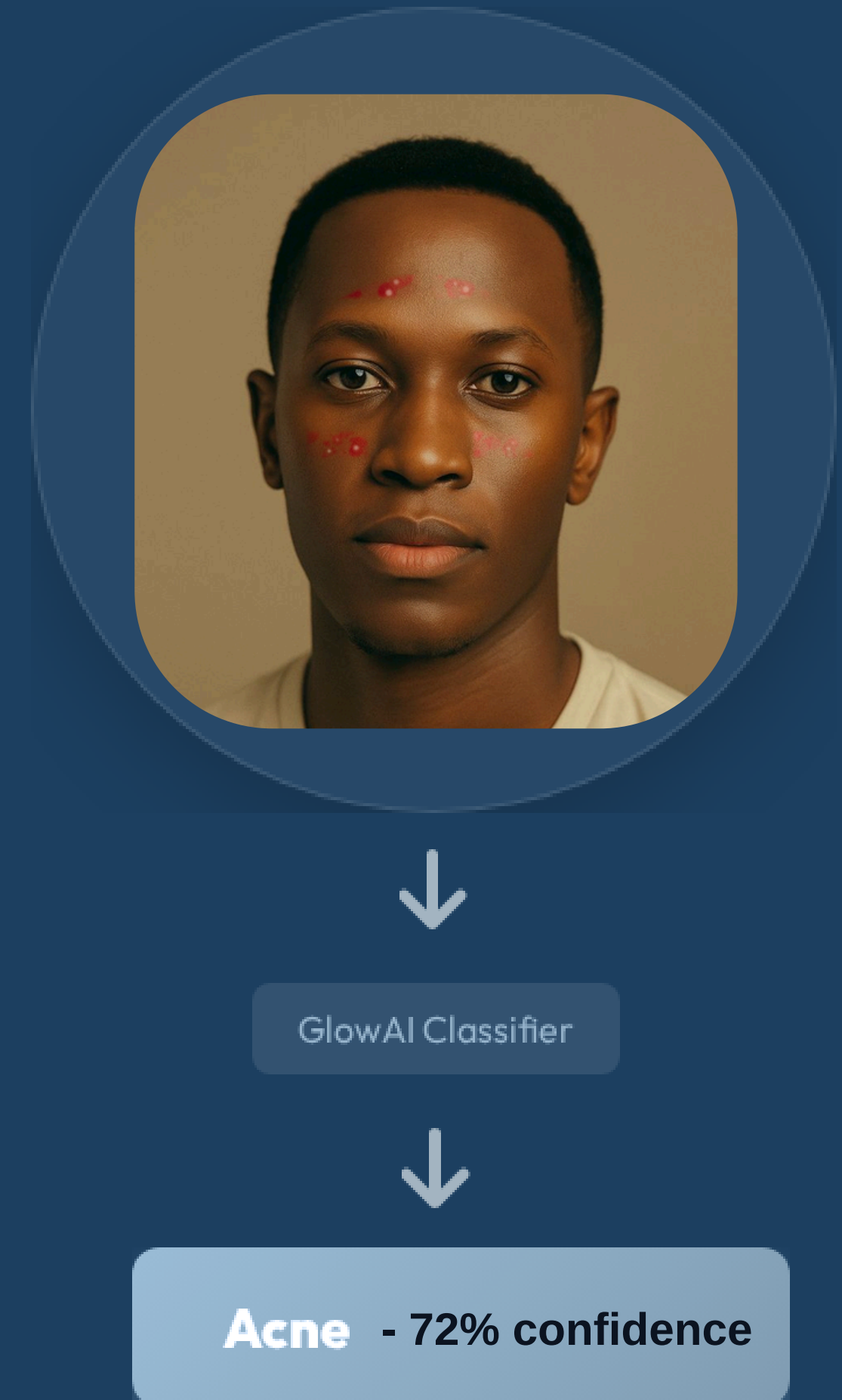
Project Definition

| Goal & Motivation

- ✓ **Goal:** Classify face images into three classes – **acne**, **erythema**, **none**.
- ✓ **Motivation:** Quick pre-screening from consumer-grade images (selfies), not a medical diagnosis.

| Input & Output

- ✓ **Input:** RGB face image (frontal selfie).
- ✓ **Output:** Predicted class + confidence score.



Project Achievements & Novelty

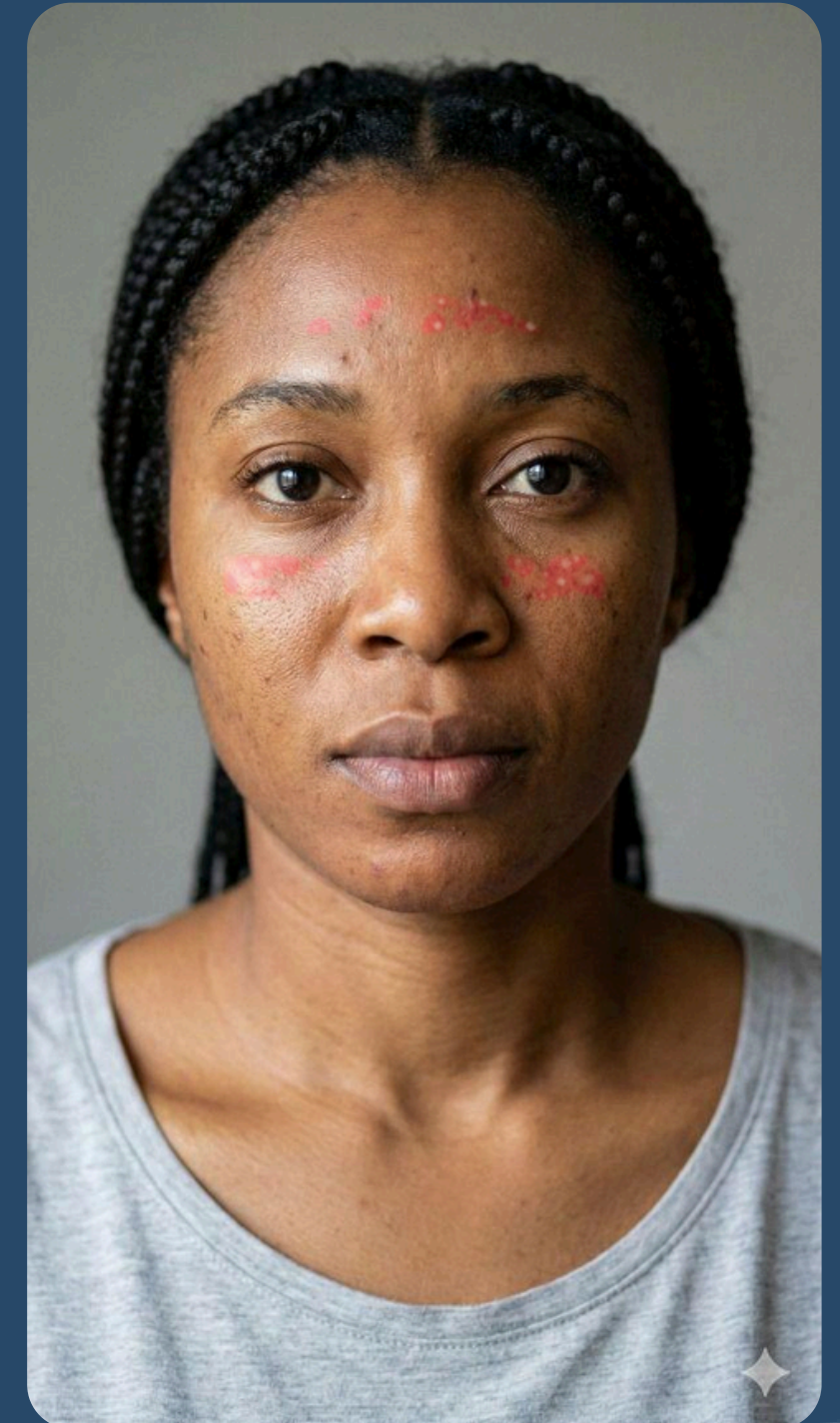
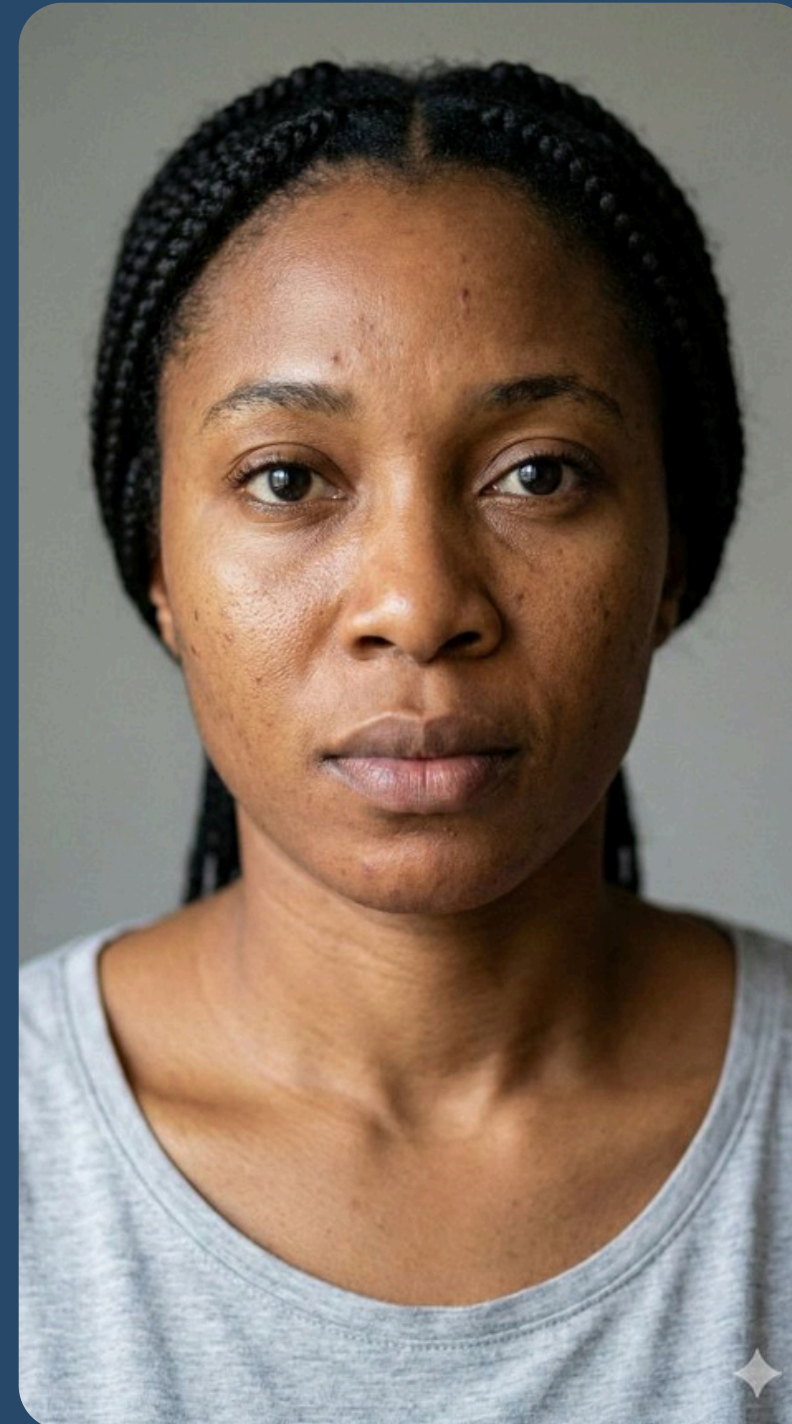
Achievements

- ✓ Trained a 3-class classifier (EfficientNet-B2) combining real and synthetic data.
- ✓ Built an end-to-end pipeline: generation → training → evaluation → feedback.

Novelty

- ✓ **Custom Skin Simulator:** A Stable Diffusion–based generative framework for realistic simulation of acne and erythema, enabling fine-grained control over lesion location, spatial spread, and severity intensity.
- ✓ **Synthetic Data Engine:** Solving scarcity of labeled medical data.
- ✓ **Feedback Loop:** Manual set to guide improvements.

Comparison: Clean vs. Synthetic

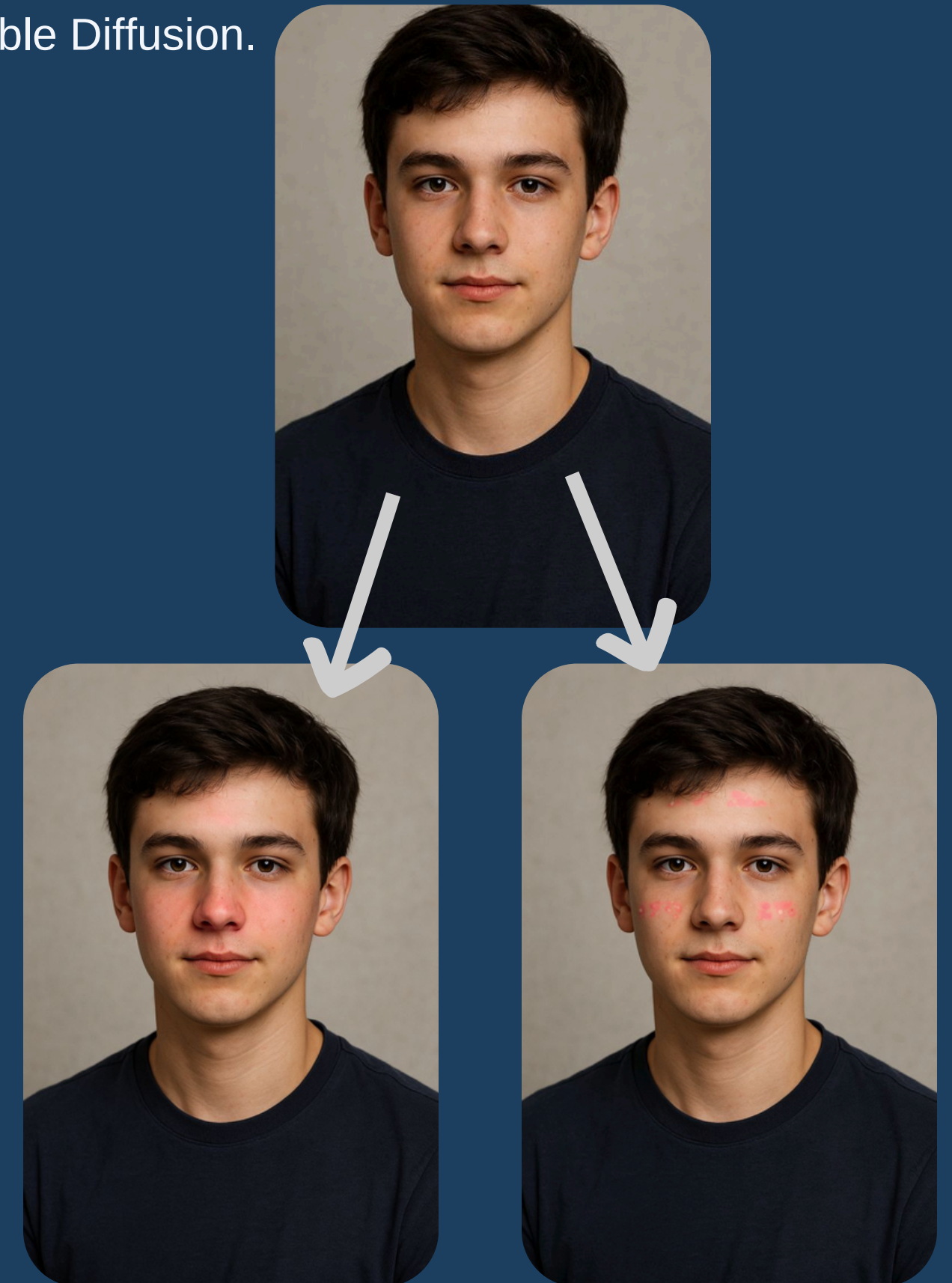


Synthetic Data Generation Using Stable Diffusion

To increase data volume and diversity, we generated synthetic facial images using Stable Diffusion.

| Data Generation Process:

- ✓ Clean facial images were used as base images.
- ✓ Carefully designed prompts were applied to synthetically add:
 - Acne (red inflamed pimples with whiteheads)
 - Facial redness (erythema)
- ✓ The identity, facial structure, and lighting of the original image were preserved to ensure realism and consistency.
- ✓ For each generated image, a corresponding pixel-level segmentation mask was created, aligned with the synthetic skin conditions.



Synthetic Data Generation – Prompt Examples

Prompt Template 1 — Acne Simulation

Generate a realistic version of the same face with mild to moderate acne vulgaris

Add red inflammatory pimples and pustules with visible whiteheads

Target areas: cheeks, forehead, chin

Acne should blend naturally with existing skin texture and lighting

Do not alter facial identity, structure, expression, hair, background, or image quality



Prompt Template 2 — Facial Redness (Erythema)

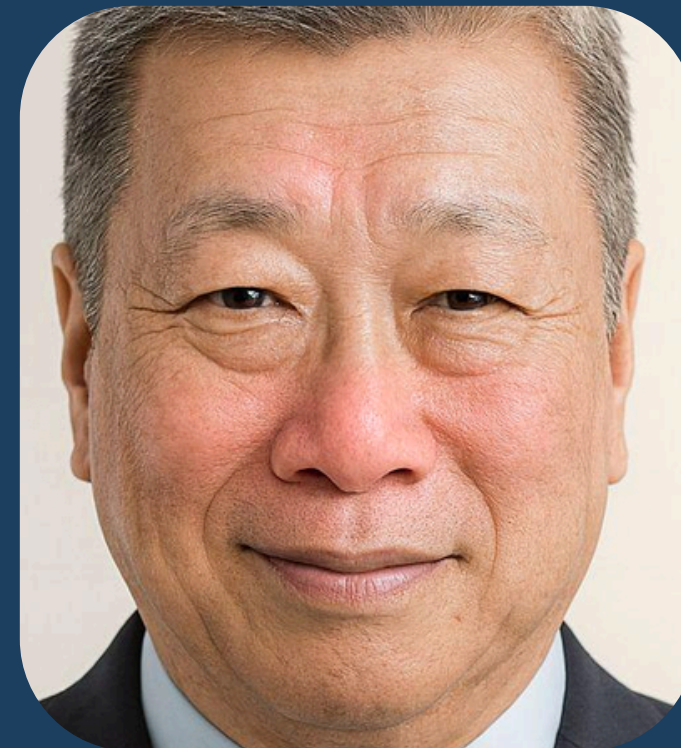
Generate a second version of the same face with realistic facial redness

Add diffuse erythema with smooth, uneven gradients

Target areas: cheeks, nose, chin

No acne, pimples, or texture changes

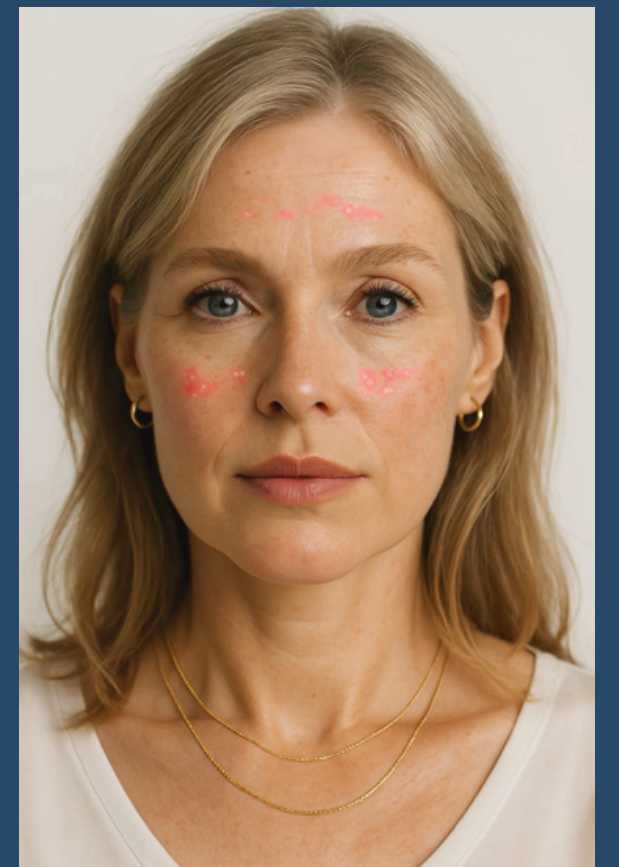
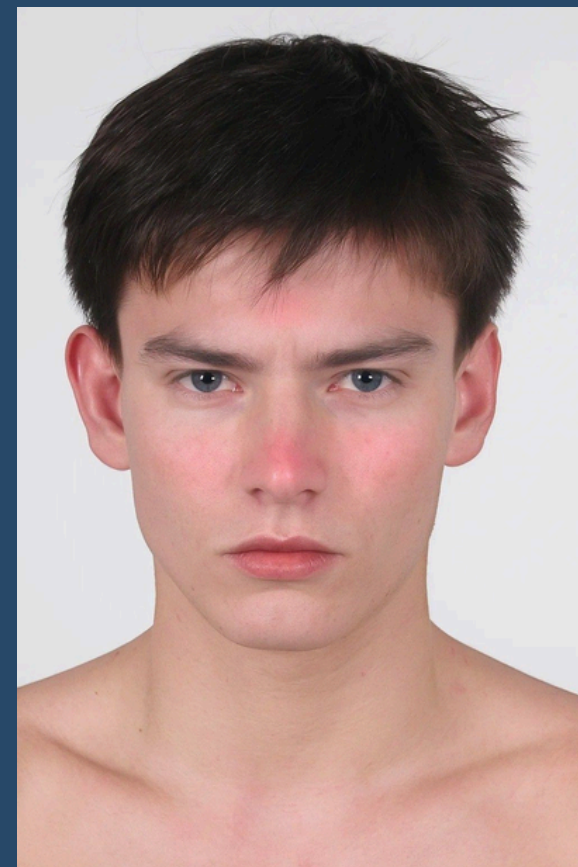
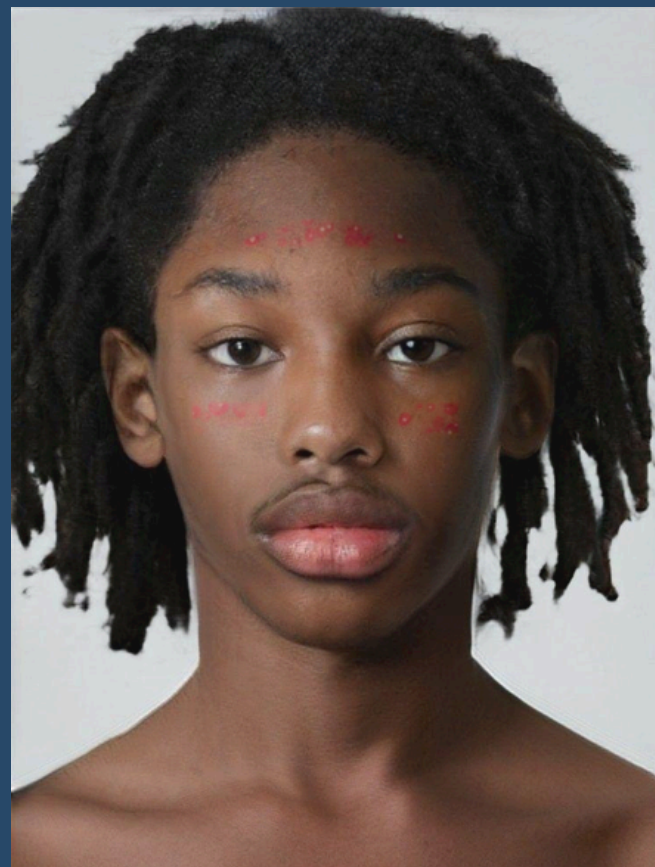
Preserve exact identity, pose, lighting, shadows, and background



BEFORE



AFTER



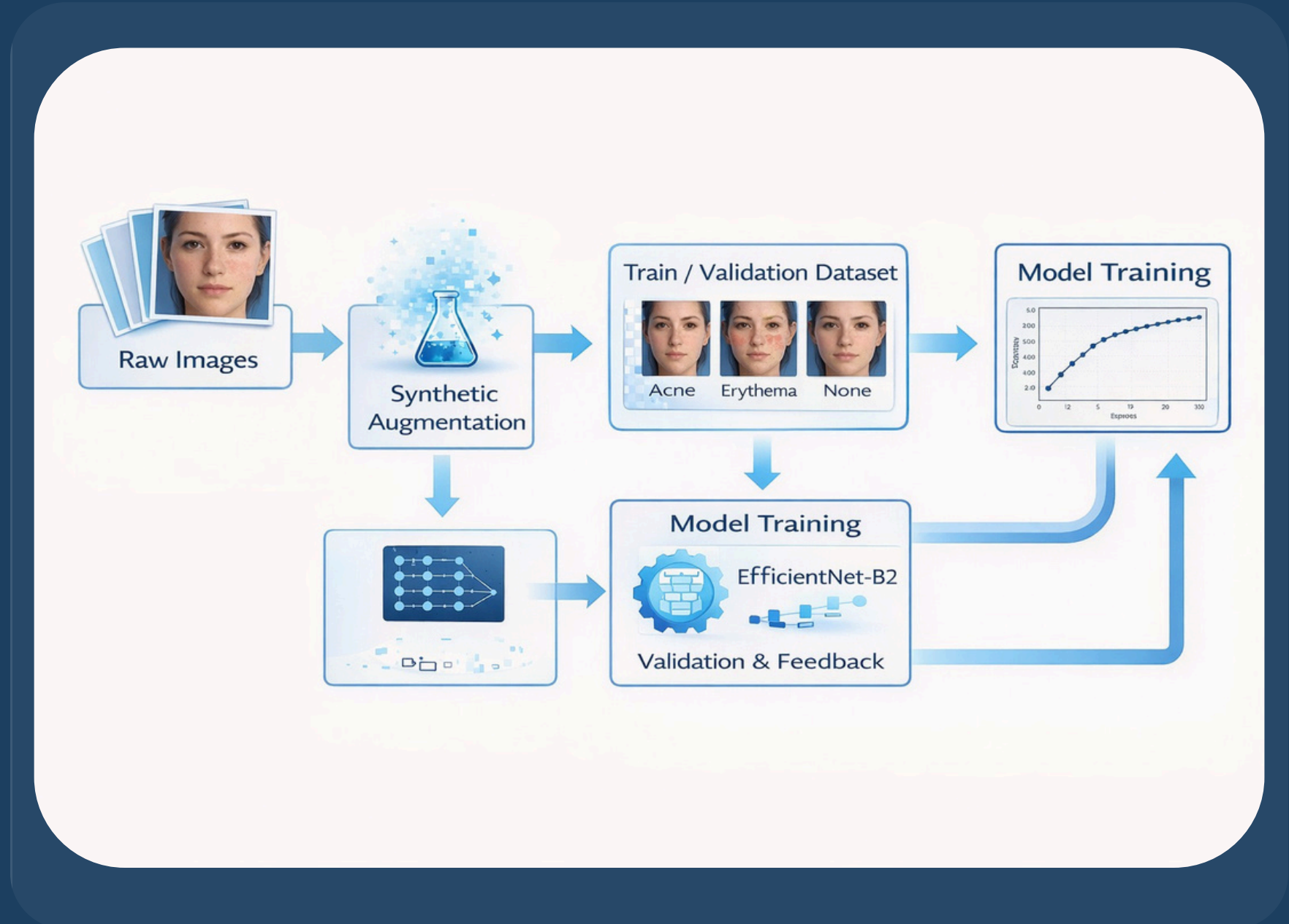
Methodology & Pipeline

Methodology

- ✓ **Data:** 3-class dataset (Acne/Erythema/None) mixed with synthetic data.
- ✓ **Preprocessing:** Resize 224×224, Normalize (ImageNet).
- ✓ **Augmentations:** Flip, rotation, color jitter (mimic selfies).
- ✓ **Training:** EfficientNet-B2, Two-stage fine-tuning, AdamW, Cross-Entropy.

Evaluation

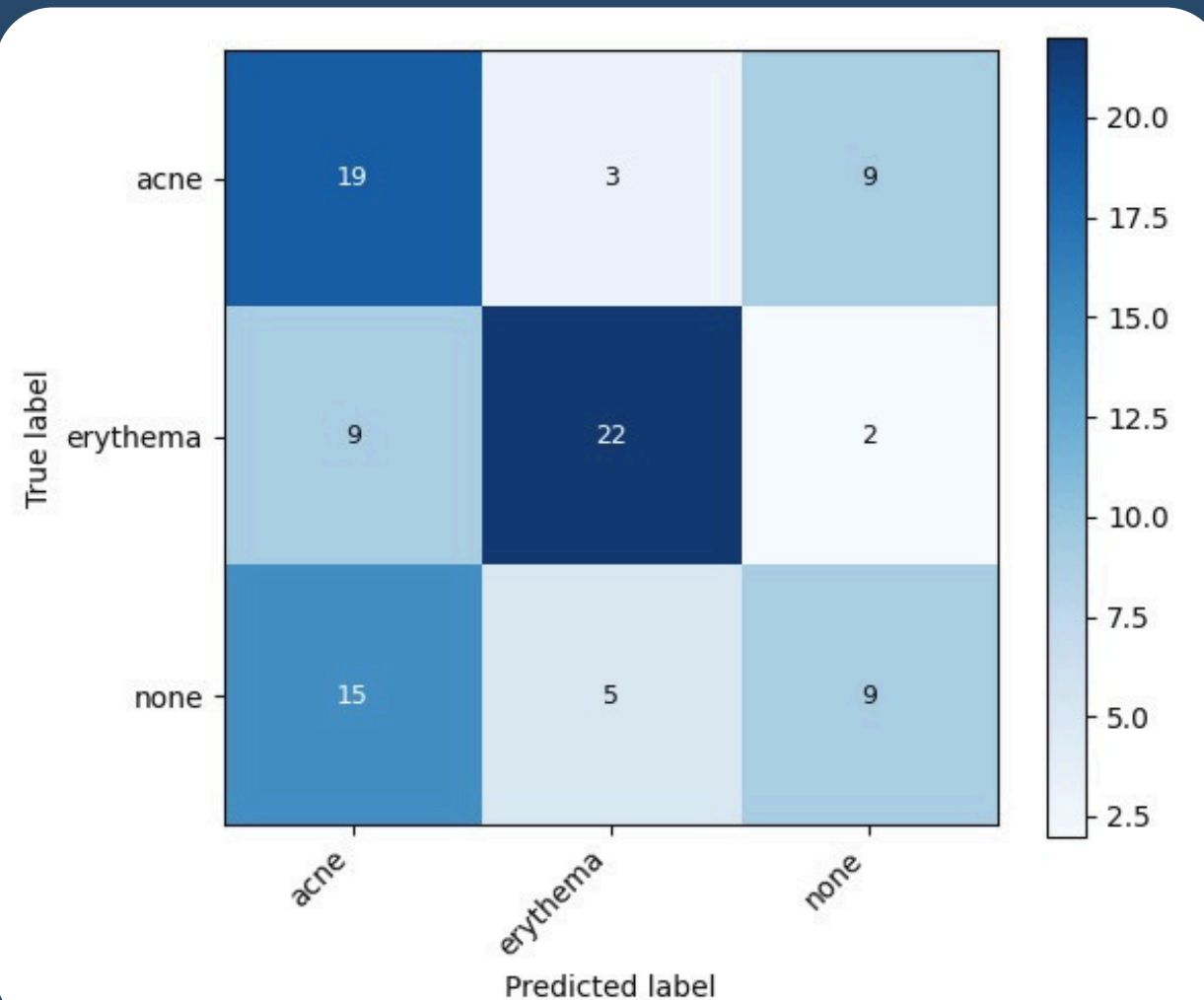
- ✓ Selected best model by validation accuracy.
- ✓ Analyzed per-class behavior via confusion matrix.



Results: Validation & Manual Test

Validation Results

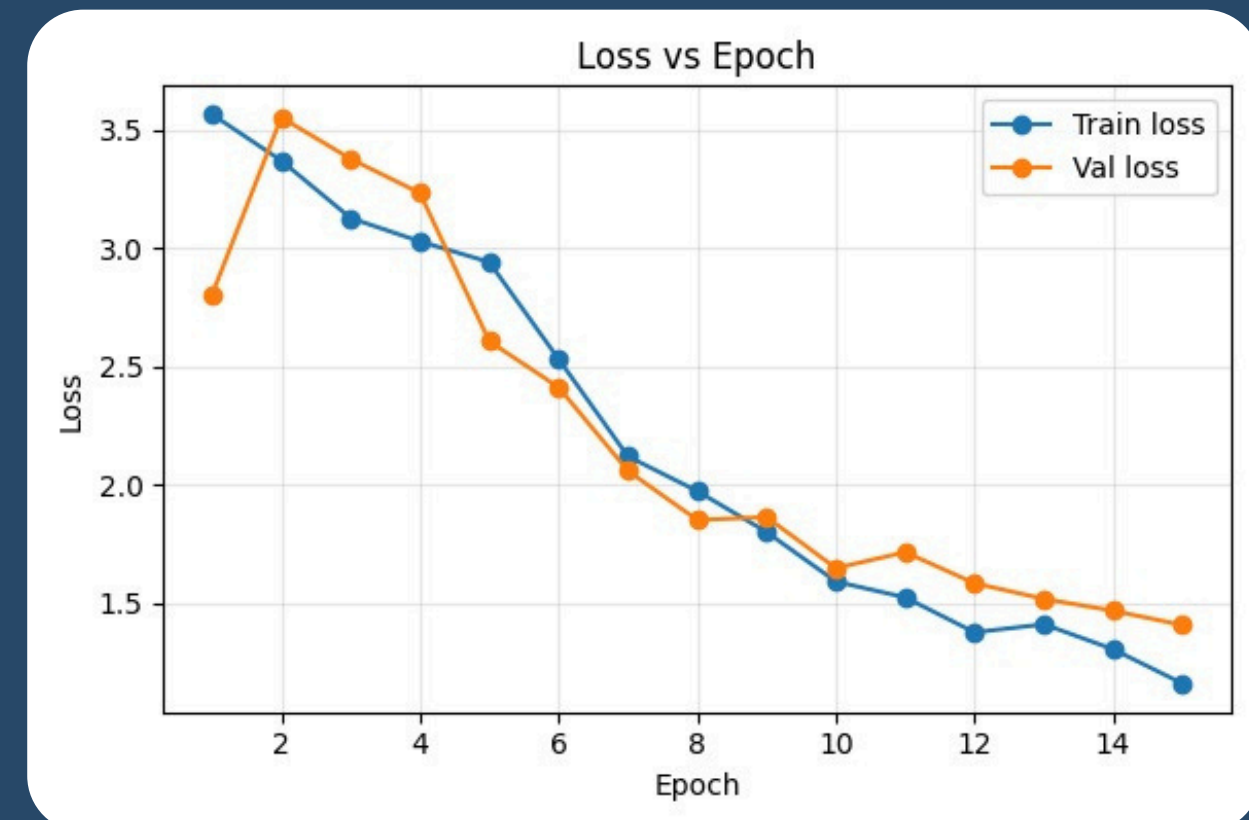
- ✓ Best Val Accuracy: ~54%.
- ✓ Main issue: "None" often predicted as "Acne/Erythema".



Confusion Matrix

Manual Test (Real Images)

- ✓ Acne: 11/15 | Erythema: 8/9 | None: 17/17
- ✓ Overall Accuracy: ~88% on curated real-world set.
- ✓ Significantly better performance on clean input than raw validation metrics suggest.



Training vs Validation Accuracy

Conclusion & Future Work

Conclusion

- ✓ Built full synthetic+real classifier pipeline.
- ✓ Reached ~**54% Val** / ~**88% Real Test** accuracy.
- ✓ Synthetic data is useful but highlights class overlap challenges.

Limitations

- ✓ Separating mild conditions is difficult.
- ✓ Class imbalance impacts robustness.
- ✓ Single-label setup is a simplification of reality.

Future Work

- ✓ Expand expert-labeled datasets.
- ✓ Move to pixel-level segmentation.
- ✓ Use diffusion models for better realism.

"These improvements could lead to a more accurate and clinically relevant skin analysis system."